

Assignment 2: The Effects of Augmented Reality Display Smart Glasses on Everyday Life

In September 2025, Meta released its first set of augmented reality (AR) display smart glasses, prompting discussions about the ethical, privacy, and surveillance implications in society. Their integration into society has rapidly advanced, although the concept of smart glasses is not new. The first generation of smart glasses by Meta was the Ray-Ban Stories. These were introduced in September 2021 and featured cameras, open-ear speakers, and microphones.¹ Highlighting the technologies integration to Meta's social apps. In October 2023, Meta upgraded these glasses with improved cameras and audio and included AI-powered features. Meta's third-generation smart glasses, were released in September of 2025 and introduced a full AR display, allowing users to overlay digital content, such as navigation, messages and captions directly into their visual field.² The latest prescription models, said to be released in April 2026, further expand these various accessibility features while making them more "wearable." With the rapid evolutions of Meta's smart glasses as wearable technologies, issues of privacy, surveillance, data protection, and the potential for self-censorship have tangible impacts on individuals' daily lives and across a variety of environments.

It is important to examine how Meta's AR smart glasses function and what makes them significantly different from previous wearable technologies. The latest models combine a forward-facing camera with an in-lens augmented reality display, allowing users to capture their surroundings and receive digital information in real time for navigation, messaging, and captioning features that can project directly into the user's field of vision. In addition to the display, the system incorporates a neural band interface that allows the user to control the glasses with subtle hand and wrist movements. The device's noticeability is reduced in everyday environments, by enabling the ability for hands-free interaction. Artificial intelligence (AI) capabilities are also available to provide visual assistance, further enhancing usability and accessibility.

The impact of Meta's AR smart glasses extends beyond technological innovation. It affects multiple groups of individuals in both positive and negative ways. As these devices are integrated into everyday life, appearing as ordinary eyewear, the subtle activation cues and small cameras make it challenging for bystanders to know when they are being filmed. The Ethical OS "Surveillance State" risk zone warns about technologies that make user monitoring invisible,

¹ "Ray-Ban and Facebook Introduce Ray-Ban Stories, First-Generation Smart Glasses." Tech at Meta, Meta, 9 Sept. 2021, tech.facebook.com/reality-labs/2021/9/ray-ban-and-facebook-introduce-ray-ban-stories-first-generation-smart-glasses/. Accessed 5 Apr. 2026.

² "Meta Ray-Ban Display: Breakthrough AI Glasses Available Now | Meta Quest Blog | Meta Store." *Meta.com*, Meta, 2025, www.meta.com/blog/meta-ray-ban-display-ai-glasses-connect-2025/?srsltid=AfmBOooNcjl9FNKQW2aZC6m0PygTNWttl1X6WzUeiunPqXXuH_MhOij. Accessed 5 Apr. 2026.

leading to social normalization.³ Meta's AR glasses accelerate this trend by turning everyday interactions into potential data collection opportunities. The lack of indicators blurs the boundaries between public and private life. Since anyone can go out and purchase a pair of these glasses, surveillance is not limited to institutions or corporations, where people often expect some lack of privacy; it becomes peer-to-peer. Whether at a grocery store, gym, park, bus, or restaurant, any person wearing these glasses becomes a mobile surveiller.

Freedom of expression is also threatened by Meta's AR glasses by creating a "chilling effect". When individuals cannot tell whether they are being recorded, they may avoid talking about controversial topics, expressing dissenting opinions, or engaging in spontaneous behavior.⁴ Free expression protects offensive or unpopular ideas because they are the most vulnerable to suppression; however, pervasive peer-to-peer surveillance undermines this protection by potentially pressuring people to self-censor in public spaces. Although AR smart glasses do not explicitly restrict speech, the fear of being captured or misinterpreted can be enough to silence expression. We have also discussed that censorship is not limited to government bans or prior restraint; it can also occur indirectly when people alter their behavior out of fear of the consequences. Meta's AR smart glasses contribute to this form of social censorship by making everyday interactions feel monitored. According to JS Mill's argument, censorship is dangerous because it can silence truths or minority perspectives that would otherwise contribute to public discourse.⁵ When people avoid certain conversations or actions because of worry a bystander may be recording, this is the result of a subtle, but powerful, form of censorship driven by technology-enabled social pressures.

For students, Meta's AR smart glasses introduce distinct risks related to academic integrity, social pressure, and the normalization of surveillance. Carleton College designated the 2024-2025 and 2025-2026 academic years to explore the benefits and challenges of AI. In an article by Brian Thevenot, he argues that while Meta's glasses offer powerful academic tools, such as real-time, translation, object recognition, and even basic coding assistance, they also raise difficult questions about how these technologies should be integrated into classroom environments.⁶ These features complicate long-standing normalcies around academic integrity, since students could discretely access information, receive AI-generated hints, or capture classroom content without detection. This article highlights the tensions between innovation and fairness to determine whether these glasses should be allowed at all, especially within higher education. In an in-class discussion, we debated the installation of a video conferencing tool for

³ Franklin, Jeff, "The Ethical OS." (PowerPoint presentation, CPRE 2340, Iowa State University, 3 Feb. 2026), slide 10.

⁴ Franklin, Jeff, "Freedom of Expression." (PowerPoint presentation, CPRE 2340, Iowa State University, 19 Feb. 2026), slide 10.

⁵ Franklin, Jeff, "Censorship & Net Neutrality." (PowerPoint presentation, CPRE 2340, Iowa State University, 26 Feb. 2026), slides 2-3.

⁶ Thevenot, Brian. "Meta AI Ray-Bans: Wearable AI and Learning?" Carleton.edu, Carleton, 6 Oct. 2025, www.carleton.edu/ai/blog/meta-ai-ray-bans-wearable-ai-and-learning/. Accessed 9 Apr. 2026.

K-12 students' laptops for virtual classes that can remotely capture images.⁷ Just as a school district may feel pressure to deploy tools with invasive capabilities because of limited resources, colleges may feel similar pressure to accept wearable AI since standardizing these technologies is not straightforward or conveniently solvable.

In professional environments, employees already navigate their workdays with monitoring systems, such as productivity-tracking systems, security cameras, and algorithmic management tools. In Janice Gassam Asare's article with Forbes, she shows how easily Meta's smart glasses can turn ordinary workplace interactions into covert data collection. She describes an incident in which an esthetician wore Meta smart glasses during a waxing appointment, revealing that customers and coworkers may be recorded without consent.⁸ Many workplaces lack established policies governing smart glasses which leave employees and clients vulnerable to misuse. Plus, these risks are not evenly distributed to all members of society. This article also notes how marginalized workers, specifically Black employees, could be disproportionately harmed by surveillance technologies. Facial recognition or behavioral analysis technologies built into these glasses can reinforce existing biases to negatively affect marginalized groups. Meta has stated that these features are restricted and are out of reach to users, but many communities and individuals have been able to share methods to bypass these limitations since facial recognition is built into multiple layers of the device.

Meta's AR smart glasses have also been said to help disabled communities. With voice prompts, the glasses can be prompted to look and describe or read users' environments, offering hands-free access to information that would otherwise require a smartphone or sighted assistance. A review from the American Foundation for the Blind notes that the glasses' object recognition, scene description, and text-reading features can meaningfully support users with low vision by helping them identify items or read printed materials. However, this review also emphasizes limitations, most likely due to being designed for content creation and social media, not accessibility.⁹ This is important to note because the intention behind the services and features of the device impact whether the device could be considered ethical. If it was intended to serve disabled communities, especially blind and deaf communities, society may be more willing to give up some of their freedoms to privacy and self-expression.

Meta has taken several measures to address the privacy and ethical concerns raised by their AR smart glasses; although, they seem limited in their effectiveness. They have claimed privacy features, such as a small LED recording indicator meant to signal when the camera is

⁷ Franklin, Jeff, "Privacy versus Security." (PowerPoint presentation, CPRE 2340, Iowa State University, 5 Mar. 2026), slide 3.

⁸ Gassam, Janice. "The Hidden Dangers of Smart Glasses: When Workplace Tech Turns into Surveillance." Forbes, 30 Sept. 2025, www.forbes.com/sites/janicegassam/2025/09/30/the-hidden-dangers-of-smart-glasses-when-workplace-tech-turns-into-surveillance/. Accessed 5 Apr. 2026.

⁹ Kelley, Steven. "A Review of the Ray-Ban Meta AI Glasses for People with Low Vision." The American Foundation for the Blind, 2025, afb.org/aw/fall2025/meta-glasses-review. Accessed 9 Apr. 2026.

active. Data-minimization controls, including options to delete stored content or restrict cloud uploads, have also been provided. Guidelines are available for respectful usage that encourages users to avoid recording in sensitive environments. However, research from numerous studies suggests that these measures fall short of meaningful privacy and data protection. A 2025 study published by NPJ Digital Medicine found that major wearable-technology companies, including Meta, consistently demonstrate weak transparency practices, unclear data-handling policies, and limited accountability mechanisms. This study claims that the broader wearable-tech ecosystem lacks adequate privacy governance, leaving users exposed to unclear or overly broad data collection.¹⁰ Similarly, the Electronic Frontier Foundation warns that Meta's smart glasses automatically upload photos and videos to the Meta AI app, and that some of this footage is reviewed for AI training by human contractors. They also note that the hardware and data pipeline already support facial recognition. This raises a larger concern about future misuse, even if the feature is not enabled.¹¹ With these articles considered, Meta has implemented surface-level privacy features, but it is not enough to fully address the structural risks associated with peer-to-peer surveillance.

Additional safeguards, given Meta's resources and technical capabilities, are both feasible and reasonable to address the concerns that follow their AR smart glasses. First, Meta could have implemented mandatory, highly visible recording indicators, rather than relying on a subtle LED light to make recording obvious to bystanders. Second, an opt-in local-storage model, where photos and videos remain on the device unless a user explicitly chooses to upload them, could have been adopted. This would greatly reduce unnecessary data aggregation and limit the risk of human review. Requiring explicit consent before any cloud transfer would also align with the best practices recommended in the NPJ Digital Medicine study. Third, no facial recognition support at any layer of the ecosystem should be enforced. Stronger technical barriers could have been implemented to prevent both official and unofficial use. This could include blocking third-party apps or preventing firmware modifications that enable facial recognition. Fourth, transparent and independent data audits could have been implemented to clearly disclose what data is being collected, how long it is stored, who can access it, and how it is used in AI training, further building accountability and public trust. Finally, better societal guidelines could have been embedded directly in the product, including onboarding modules to teach users about consent, bystander privacy, and responsible recording practices.

As wearable technologies develop and quickly become embedded in everyday life, they will reshape how people move through public and private spaces. Meta's AR glasses reveal how

¹⁰ Doherty, Cailbhe, et al. "Privacy in Consumer Wearable Technologies: A Living Systematic Analysis of Data Policies across Leading Manufacturers." NPJ Digital Medicine, vol. 8, no. 1, 14 June 2025, [www.nature.com/articles/s41746-025-01757-1](https://doi.org/10.1038/s41746-025-01757-1), <https://doi.org/10.1038/s41746-025-01757-1>. Accessed 5 Apr. 2026.

¹¹ Klosowski, Thorin. "Think Twice before Buying or Using Meta's Ray-Bans." Electronic Frontier Foundation, 10 Mar. 2026, www.eff.org/deeplinks/2026/03/think-twice-buying-or-using-metas-ray-bans. Accessed 9 Apr. 2026.

even a single device, worn by even one person, can influence the privacy, autonomy, and comfort of a variety of groups around them. They challenge people's rights to privacy, censorship, freedom of expression, and align with multiple risk zones from the Ethical OS Model. Companies, policymakers, and communities must recognize the broad social consequences these devices introduce, and push for responsible design and governance to keep everyone protected. The future of wearable technologies, like Meta's growing AR smart glasses, will have to balance rapid innovation with the fundamental rights of the people who will live alongside these devices.